

LECTURE NOTES MATH255

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1. IMPORTANT DEFINITIONS AND THEOREMS

In these notes we collect important theorems and definitions that we have used throughout the course.

1.1. Basic definitions and properties of rings and fields.

Definition 1.1 (Definition of a ring). *A ring R is a set with two binary operations $+$, $\cdot$ such that $(R, +)$ is an abelian group, $\cdot$ is associative:*

$$(1.1) \quad (a \cdot b) \cdot c = a \cdot (b \cdot c), \quad a, b, c \in R$$

and $\cdot$ distributes over addition:

$$(1.2) \quad c \cdot (a + b) = c \cdot a + c \cdot b, \quad (a + b) \cdot c = a \cdot c + b \cdot c, \quad a, b, c \in R.$$

Definition 1.2 (Definition of a field). *A field F is a commutative ring with unity $1 \neq 0$ such that every $a \in F \setminus \{0\}$ is a unit.*

Definition 1.3 (Definition of a subring). *Let R be a ring. We say that a subset $S \subset R$ is a subring if S is closed under $+$ and $\cdot$, and if S forms a ring with the operations $+$ and $\cdot$.*

Definition 1.4 (Homomorphisms of rings). *Let R and S be rings. We say that $\phi : R \rightarrow S$ is a ring homomorphism (or just a homomorphism) if:*

- (i) $\phi(a + b) = \phi(a) + \phi(b)$ for all $a, b \in R$,
- (ii) $\phi(ab) = \phi(a)\phi(b)$ for all $a, b \in R$.

Definition 1.5 (Isomorphisms of rings). *Let R and S be rings. We say that $\phi : R \rightarrow S$ is an isomorphism if ϕ is a bijective homomorphism.*

Definition 1.6 (Homomorphisms of fields). *Let F and L be fields. We say that $\phi : F \rightarrow L$ is a field homomorphism (or just a homomorphism) if:*

- (i) $\phi(a + b) = \phi(a) + \phi(b)$ for all $a, b \in F$,
- (ii) $\phi(ab) = \phi(a)\phi(b)$ for all $a, b \in F$,
- (iii) $\phi(1_F) = 1_L$.

Definition 1.7 (Isomorphisms of fields). *Let F and L be fields. We say that $\phi : F \rightarrow L$ is an isomorphism if ϕ is a bijective homomorphism.*

Definition 1.8 (Zero divisors). *Let R be a ring. If $a, b \in R$ are non-zero, $a, b \neq 0$, and satisfy $ab = 0$ then we say that a and b are zero divisors.*

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Definition 1.9 (Integral domains). *Let R be a ring. We say that R is an integral domain if R is a commutative ring with unity $1 \neq 0$, and R has no zero divisors.*

Theorem 1.1 (Fermat's little theorem). *Let p be a prime and let $a \in \mathbb{Z}$ be any integer that is not divisible by p . Then $a^{p-1} - 1$ is divisible by p (or: $a^{p-1} = 1 \pmod{p}$).*

Theorem 1.2 (Euler's generalization of Fermat's little theorem). *Let $n \geq 1$ be any positive integer and $a \in \mathbb{Z}$ be any integer that is relatively prime to n . Then $a^{\varphi(n)} - 1$ is divisible by n where $\varphi : \mathbb{N} \rightarrow \mathbb{N}$ is the Euler totient function (or: $a^{\varphi(n)} = 1 \pmod{n}$).*

Definition 1.10 (Fraction fields). *Let R be an integral domain. A field F is a fraction field for R if there exists an injective ring homomorphism $\psi : R \rightarrow F$ such that if we write $D = \text{Im}(\psi)$ then any $x \in F$ can be written as:*

$$(1.3) \quad x = \frac{a}{b}$$

with $a \in D$ and $b \in D \setminus \{0\}$.

Theorem 1.3 (Fraction field theorem). *Let R be an integral domain, then R has a fraction field F . Moreover, the fraction field of R is unique up to isomorphism.*

1.2. Polynomials.

Definition 1.11 (Definition of a polynomial). *Let R be a ring. A polynomial with coefficients in R in the variable x is a formal sum:*

$$(1.4) \quad f(x) = \sum_{j=0}^{\infty} a_j x^j$$

where $a_j \in R$ and $a_j \neq 0$ for finitely many j . The polynomial ring with coefficients in R and unknown x is the set of all polynomials with coefficients in R and unknown x , denoted $R[x]$.

Theorem 1.4 (Division algorithm for polynomials). *Let F be a field, and $F[x]$ its polynomial ring. If $f(x), g(x) \in F[x]$ then there are unique polynomials $q(x), r(x) \in F[x]$ such that $f(x) = q(x)g(x) + r(x)$ where $\deg(r(x)) < \deg(g(x))$.*

Definition 1.12 (Irreducible polynomials). *A non-constant polynomial $f(x) \in F[x]$ is irreducible over F if its only factors are multiples of $f(x)$ and constants $a \in F$. Equivalently, $f(x)$ is irreducible over F if $f(x)$ can not be written as $f(x) = g(x)h(x)$ where both $g(x)$ and $h(x)$ have strictly lower degree than $f(x)$.*

Theorem 1.5 (Eisenstein's criterion). *Let $a_n x^n + \dots + a_1 x + a_0 = f(x) \in \mathbb{Z}[x]$ be a polynomial with integer coefficients. If there is a prime p that does not divide a_n , but divides each $a_{n-1}, \dots, a_0$, and a_0 is not divisible by p^2 , then $f(x)$ is irreducible.*

Theorem 1.6 (Unique factorization for polynomials). *Let $f(x) \in F[x]$ be a non-constant polynomial with coefficients in some field F . There are irreducible polynomials $p_1(x), \dots, p_n(x)$ such that $f(x) = p_1(x) \dots p_n(x)$. Moreover, the irreducible polynomials $p_1(x), \dots, p_n(x)$ are unique up to changing the order of the polynomials and multiplying them with units. That is, if $f(x) = q_1(x) \dots q_m(x)$, with each $q_j(x)$ irreducible, then $n = m$ and after changing the order of the $q_j(x)$ we find some constants $a_j \in F \setminus \{0\}$ such that $q_j(x) = a_j p_j(x)$.*

Theorem 1.7 (Existence of zeros of polynomials). *Let $f(x) \in F[x]$ be a non-constant polynomial with coefficients in F . There exists a finite field extension E/F and some $\alpha \in E$ such that $f(\alpha) = 0$. That is, any non-constant polynomial has a zero (in some field extension).*

1.3. Ideals and quotients.

Definition 1.13 (Definition of ideals). *Let R be a ring. An additive subgroup $I \subset R$ is an ideal if:*

$$(1.5) \quad aI \subset I, \quad Ia \subset I$$

for all $a \in R$ where:

$$(1.6) \quad aI = \{ax \mid x \in I\}, \quad Ia = \{xa \mid x \in I\}.$$

Definition 1.14 (Definition of a quotient ring). *Let R be a ring and $I \subset R$ an ideal. We define the quotient ring of R by I as the quotient $R/I = \{x + I \mid x \in R\}$.*

Theorem 1.8 (Fundamental homomorphism theorem for rings). *Let $\phi : R \rightarrow S$ be a homomorphism of rings with kernel $I = \ker \phi$. Then there exists an isomorphism $\mu : R/I \rightarrow \phi(R)$ that satisfies $\mu \circ \gamma = \phi$ where $\gamma : R \rightarrow R/I$ is the quotient map.*

Definition 1.15 (Maximal ideals). *Let R be a ring. An ideal $M \subset R$ is a maximal ideal if M is a proper ideal (i.e., $M \neq R$) such that M is not properly contained in any proper ideal (i.e., there is no ideal $M' \subset R$ such that $M' \neq R$ and $M \subsetneq M'$).*

Definition 1.16 (Prime ideals). *Let R be a ring. An ideal $P \subset R$ is a prime ideal if $ab \in P$ implies $a \in P$ or $b \in P$.*

1.4. Field extensions.

Definition 1.17 (Algebraic elements). *Let E/F be a field extension of F . We say that $\alpha \in E$ is algebraic over F if there exists a polynomial $f(x) \in F[x]$ such that $f(\alpha) = 0$.*

Definition 1.18 (Finite extensions). *Let E/F be an extension of F . We say that E is a finite extension of F if the dimension of E as a F -vector space is finite. We write the dimension of E as a F -vector space as $[E : F]$.*

Theorem 1.9 (Tower property). *If L/F and E/L are finite extensions, then E is a finite extension of F and $[E : F] = [L : F] \cdot [E : L]$.*

Definition 1.19.

Definition 1.20 (Algebraically closed fields). *A field F is algebraically closed if every polynomial $f(x) \in F[x]$ has a zero in F .*

Definition 1.21 (Algebraic closures). *Let F be a field. An algebraic closure of F is an algebraic extension of F , $\overline{F}/F$, such that $\overline{F}$ is algebraically closed.*

Theorem 1.10 (Existence of algebraic closures). *Every field F has an algebraic closure $\overline{F}$. Moreover, the algebraic closure of F is unique up to isomorphism.*

1.5. Various types of domains.

Definition 1.22 (Unique factorization domains). *Let R be an integral domain. We say that R is a unique factorization domain (UFD) if the following two conditions hold:*

- (i) *every $a \in R$ that is not 0 and not a unit can be written as a product of irreducible elements in R ,*
- (ii) *if $a = p_1 \dots p_n = q_1 \dots q_m$, p_j and q_i all irreducible in R , then $n = m$ and after reordering the elements we have that p_j and q_j are associates.*

Definition 1.23 (Principal ideal domains). *An integral domain R is a principal ideal domain (PID) if every ideal $I \subset R$ is a principal ideal. That is, if $I \subset R$ is an ideal then there is some $a \in R$ such that $\langle a \rangle = I$.*

Theorem 1.11 (Ascending chain condition). *Let R be a PID. If $I_1 \subset I_2 \subset I_3 \subset \dots$ are nested ideals in R then there is some $j < \infty$ such that for every $k \geq 0$ we have $I_{j+k} = I_j$. That is, any nested sequence of ideals terminates at some I_j .*

Theorem 1.12 (PIDs are UFDs). *Let R be a PID. Then R is a UFD.*

Definition 1.24 (Euclidean norms). *Let R be an integral domain. A Euclidean norm on R is a function $\nu : R \setminus \{0\} \rightarrow \mathbb{N}_0$ such that:*

- (i) *If $a, b \in R \setminus \{0\}$ then there exists $q, r \in R$ such that $a = bq + r$ and either $r = 0$ or $\nu(r) < \nu(b)$.*
- (ii) *If $a, b \in R \setminus \{0\}$ then $\nu(ab) \geq \nu(a)$.*

Definition 1.25 (Euclidean domains). *Let R be an integral domain. We say that R is a Euclidean domain if there is a Euclidean norm ν defined on R .*

Theorem 1.13 (Euclidean domains are PID). *Let R be a Euclidean domain. Then R is a PID.*

Definition 1.26 (Multiplicative norms). *Let R be an integral domain. A map $N : R \rightarrow \mathbb{Z}$ is a multiplicative norm if:*

- (i) *$N(a) = 0$ if and only if $a = 0$,*
- (ii) *$N(ab) = N(a)N(b)$ for all $a, b \in R$.*

1.6. Galois theory.

Definition 1.27 (Conjugate numbers). *Let K be a number field. Two algebraic numbers $\alpha, \beta \in \overline{\mathbb{Q}}$ are conjugate over K if they are zeros of the same irreducible polynomial $f(x) \in K[x]$.*

Definition 1.28 (Finite normal extensions). *Let K be a number field and E/K an extension of K . We say that E is a finite normal extension of K if E is a finite extension of K , and if every irreducible polynomial $f(x) \in K[x]$ that has at least one zero in E splits in E . Equivalently, if $\alpha \in E$ then every conjugate of α over K is also in E .*

Definition 1.29 (Galois groups). *Let E/K be a finite normal extension of a number field K . We define the Galois group of the extension E/K as $\text{Gal}(E/K) = \text{Aut}(E/K)$. That is, the Galois group of E/K is the group of automorphisms of E that fix K .*

Theorem 1.14 (Primitive element theorem). *Let K be a number field and E/K a finite extension of K . There exists some $\theta \in E$ such that $E = K(\theta)$.*

Theorem 1.15 (Fundamental theorem of Galois theory). *Let K be a number field and E/K a finite normal extension of K . The following holds:*

- (i) *For any subfield $K \leq L \leq E$ there is a subgroup $H \leq \text{Gal}(E/K)$ such that $L = E^H$.*
- (ii) *For any subgroup $H \leq \text{Gal}(E/K)$ there is some subfield $K \leq L \leq E$ such that $H = H_L$.*
- (iii) *The correspondence between subgroups and subfields are inverses: $H_{E^H} = H$, and $E^{H_L} = L$.*
- (iv) *A subfield $K \leq L \leq E$ is a finite normal extension of K if and only if the corresponding subgroup $H_L \leq \text{Gal}(E/K)$ is a normal subgroup of the Galois group.*
- (v) *If $K \leq L \leq E$ is a finite normal extension of K then:*

$$(1.7) \quad \text{Gal}(L/K) \cong \text{Gal}(E/K)/H_L$$

where the quotient $\text{Gal}(E/K)/H_L$ is a group since H_L is a normal subgroup by (iv).

Definition 1.30 (Algebraic numbers solvable by radicals). *Let $\alpha \in \overline{\mathbb{Q}}$ be an algebraic number. We say that α is solvable by radicals if there are numbers $\alpha_1, \dots, \alpha_m \in \overline{\mathbb{Q}}$ and positive integers $n_1, \dots, n_m$ such that $\alpha_1^{n_1} \in \mathbb{Q}$, for $j < n$ we have $\alpha_{j+1}^{n_{j+1}} \in \mathbb{Q}(\alpha_1, \dots, \alpha_j)$, and $\alpha \in \mathbb{Q}(\alpha_1, \dots, \alpha_n)$.*

Theorem 1.16 (Galois group of algebraic numbers solvable by radicals). *Let $\alpha \in \overline{\mathbb{Q}}$ be an algebraic number that is solvable by radicals. If $f_\alpha(x) \in \mathbb{Q}[x]$ is the irreducible polynomial for α and E_α is the splitting field of $f_\alpha(x)$ then $\text{Gal}(E_\alpha/\mathbb{Q})$ is a solvable group.*

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